ROBERT J. PARK
ENVIRONMENT · CARBON · PUBLIC AFFAIRS
NORTHERN IRELAND

Environment · Carbon · Public Affairs · Northern Ireland

BRIEFING · INSTITUTIONS · SCIENCE · CARBON · NOVEMBER 2026

Faith, reason and the common home

The Catholic Church's record in science, business and the environment: what it has done, the evidence for it, how big its footprint is, and which Vatican offices now carry the work.

1.42bn
Catholics worldwide, 17.8% of the world (mid-2024)

64m
pupils in 230,924 Catholic schools, plus 4.1m university students

101,110
Catholic health and charity centres, including 5,397 hospitals

0.44 km²
Vatican City State: 887 residents, about 6,000 daily workers

15,890 t
Vatican CO_2e in 2022, 18% below 2011; 71% from vehicles

13,600+
participants in the Laudato Si' Action Platform after five years

Robert J. Park · robertjpark.co.uk · robert@robertjpark.co.uk · WhatsApp @robertjpark
Figures checked against the sources listed at the end, as at 9 November 2026. This briefing is general information, not legal, financial or investment advice.

Summary

> **Why a carbon consultant is writing about the Church.** Few institutions have run schools, hospitals and observatories for centuries, and fewer still publish an emissions inventory for a state of 0.44 km^2. This briefing looks at the record as evidence: what was done, when, and what the sources show. It is not a statement of faith, and the same approach would suit any long-lived institution.

- **Science.** The Church helped build modern science, not only tolerate it. Our calendar came from a papal reform worked out by Jesuit astronomers in 1582. A friar founded genetics, a bishop helped found geology and a priest proposed what became the Big Bang. The Vatican still runs one of the world's oldest working observatories, with a research telescope in Arizona, and an Academy of Sciences whose members have included Planck, Bohr and Schrödinger.
- **Business.** A Franciscan friar published the first printed account of double-entry bookkeeping in 1494. Rerum novarum (1891) made the case for a living wage and trade unions. A Basque priest founded Mondragon, now one of the world's largest groups of worker co-operatives. In 2022 the Vatican published guidance on faith-consistent investing.
- **Environment.** Laudato si' (2015) put care for creation at the centre of Catholic teaching. The Holy See joined the UN climate convention and the Paris Agreement in 2022 and has filed two emissions targets. More than 13,600 participants have joined the Vatican's Laudato Si' Action Platform. The Irish bishops' conference moved its investments out of fossil fuels in 2018.
- **Footprint.** 1.42 billion Catholics, about 64 million pupils in Catholic schools, 4.1 million university students and just over 101,000 health and charity centres, including 5,397 hospitals.
- **The Vatican itself.** A state of 0.44 km^2 that emitted about 15,900 tonnes of CO_2e in 2022, roughly 18% below 2011. It has pledged a 28% cut by 2035 and net zero by 2050, and is building an agrivoltaic plant outside Rome to supply all its electricity.

1. Science

The calendar on your wall

In 1582 Pope Gregory XIII promulgated the calendar the world still uses. It was developed from astronomical data, and the Jesuit mathematician Christopher Clavius was a key figure in the reform. The Vatican Observatory dates its roots to that work.

Priests who founded sciences

- **Nicolas Steno (1638–1686)**, a Danish anatomist who became a Catholic bishop and is now beatified, is counted among the founders of geology.
- **Fr Angelo Secchi (1818–1878)**, a Jesuit, helped launch astrophysics through his classification of stellar spectra.
- **Gregor Mendel (1822–1884)**, an Augustinian friar and later abbot of St Thomas's Abbey in Brno, cultivated some 28,000 plants, mostly peas, in the abbey garden between 1856 and 1863. He presented the results to the Brno Natural History Society in February and March 1865 and published them in 1866. The paper was barely cited until 1900. It is now the founding text of genetics.
- **Mgr Georges Lemaître (1894–1966)**, a Belgian priest and physicist, proposed in 1927 that the universe is expanding and later that it began from a "primeval atom". That is the model we now call the Big Bang.

Institutions that have lasted

The Pontifical Academy of Sciences traces its roots to the Accademia dei Lincei, founded in Rome in 1603 by Federico Cesi under the patronage of Pope Clement VIII. Galileo was a member. Pius IX re-founded it in 1847 and it took its present name in 1936. It has about 80 members appointed for life, chosen without regard to religion. Past members include Rutherford, Planck, Hahn, Bohr, Schrödinger and Townes, and many were appointed before they won their Nobel prizes.

The Vatican Observatory was re-founded by Leo XIII in 1891, partly to answer the charge that the Church was hostile to science. Its astronomers took part in the Carte du Ciel, the first international project to photograph and chart the whole sky. Nuns from the Sisters of the Holy Child Mary were recruited as "computers" to record the stars' positions, and those plates are now digitised. Since 1993 the Observatory has run the 1.8-metre Vatican Advanced Technology Telescope on Mount Graham, Arizona. Its primary mirror was the first made with the University of Arizona's spin-casting furnace technique.

> **A fair record.** The Galileo case is the exception everyone remembers. Galileo was condemned in 1633. In 1979 John Paul II asked for the case to be re-examined, and on 31 October 1992, after the commission reported, he told the Academy of Sciences that the errors of Galileo's judges must be recognised. An institution that admits its mistakes, even late, is easier to trust on the rest of its record.

---

2. Business and economics

The friar who wrote down double-entry bookkeeping

Luca Pacioli, a Franciscan friar and mathematician who worked with Leonardo da Vinci, published the *Summa de arithmetica* in Venice in 1494. It contained the first printed description of the double-entry method that Venetian merchants used. Pacioli did not invent it, but he set it out in full, and his book spread it across Europe. Every set of company accounts, and every carbon ledger built on the same logic, owes something to him.

A living wage and the right to organise

Leo XIII's *Rerum novarum* of 15 May 1891 is the founding document of modern Catholic social teaching. Written during the first industrial revolution, it rejected both unrestricted capitalism and socialism. It defended private property and also argued for a wage that can support a worker and their family, and for workers' right to form unions. Catholic trade unions followed across Europe and the Americas. Pius XI's *Quadragesimo anno* (1931) made explicit the principle of subsidiarity: decisions should be taken at the lowest level able to take them well. That principle runs through EU law today.

Leo XIV chose his name partly to honour Leo XIII. His first encyclical, *Magnifica humanitas*, signed on the 135th anniversary of *Rerum novarum* and published on 25 May 2026, carries the same tradition into the age of artificial intelligence. It argues that AI must serve people and the dignity of work, and calls for robust regulation.

Mondragon: a priest's co-operative

Fr José María Arizmendiarrieta arrived in the Basque town of Mondragón in 1941, in the poverty that followed the Spanish Civil War. In 1943 he founded a technical school. In 1956 five of its graduates set up a worker co-operative, Ulgor, which became Fagor. In 1959 he and his team founded a co-operative bank, Caja Laboral, to finance more of them. The Mondragon group now has about 70,000 workers in some 80 co-operatives and is one of the leading forces in the Basque economy. Each worker-owner holds one share and one vote.

Investing in line with values

In November 2022 the Pontifical Academy of Social Sciences published *Mensuram Bonam* (Good Measure), guidance for Catholic investors on bringing Church teaching into investment policy. It suggests three ways to act: engage with companies, invest in what does good, and exclude what does not. It says ESG investing is a useful starting point but not

the same thing as Catholic teaching, and that investing in line with faith need not conflict with fiduciary duty. It is guidance, not binding law, and it describes itself as a starting point. Several bishops' conferences have their own investment guidelines.

The Church as an economic actor

The Church is one of the world's largest providers of schooling and health care (section 4). Those institutions employ millions, buy goods and services, and run very large estates of buildings. That gives the Church more weight as a buyer and investor than its own small headquarters would suggest.

---

3. The environment

Teaching

Concern for creation runs back to St Francis of Assisi. Paul VI and John Paul II both wrote about it, and Benedict XVI was sometimes called the "green pope". The decisive text is Pope Francis's encyclical *Laudato si'* of 2015. It treats climate change, biodiversity loss and pollution as moral questions bound up with poverty ("integral ecology"). It calls for fossil fuels to be replaced and renewable energy developed without delay. *Laudate Deum* (2023) followed it up ahead of COP28. Leo XIV has continued the work: in July 2025 he celebrated the Church's new Mass for the Care of Creation, and in September 2025 he opened Borgo Laudato Si', an ecology training centre on the papal estate at Castel Gandolfo.

Science behind the teaching

In April 2015, two months before *Laudato si'*, the Pontifical Academies of Sciences and Social Sciences published *Climate Change and the Common Good*. Its authors included the Cambridge economist Partha Dasgupta and the climate scientist Veerabhadran Ramanathan. A declaration signed at the Vatican that month by religious, political and business leaders and scientists stated that human-induced climate change is a scientific reality and that cutting it decisively is a moral and religious imperative.

From words to commitments

- **Treaty party.** On 4 July 2022 the Holy See joined the UN Framework Convention on Climate Change on behalf of Vatican City State, and it is also a party to the Paris Agreement. It filed its first emissions target (NDC) in May 2023 and a second in September 2025 (section 6).
- **Laudato Si' Action Platform.** Launched in 2021 by the Dicastery for Promoting Integral Human Development, the platform takes dioceses, parishes, schools, universities, hospitals, religious orders, businesses and families through a seven-year plan towards "total sustainability". Its fifth annual report (November 2026) counts more than 13,600 participants and aims for 50,000 by 2031.
- **Divestment.** On 24 August 2018, the eve of Pope Francis's visit to Ireland, the all-island Irish Catholic Bishops' Conference committed to divest from the top 200 oil and gas companies by reserves within five years. It was the second bishops' conference to do so, after Belgium's. Trócaire, the Irish bishops' overseas aid agency founded in 1973, made the case to the bishops' finance committee.

---

4. The Church's footprint worldwide

These are the Church's own figures, compiled by the Vatican's Central Office of Church Statistics and published by Agenzia Fides on 16 October 2026. Population figures are as at 30 June 2024; the rest are as at 31 December 2024.

TABLE 1. THE CATHOLIC CHURCH IN NUMBERS

Measure	Figure
Catholics	1,422,144,000
Share of world population	17.8%
Bishops	5,525
Priests	407,421
Women religious	580,131
Major seminarians	103,604

TABLE 2. EDUCATION

Type	Schools	Pupils
Kindergartens	75,448	7,480,373
Primary	103,666	36,314,935
Secondary	51,810	20,340,808
Total	**230,924**	**64,136,116**
University students	–	4,091,087

TABLE 3. HEALTH AND CHARITY

Type	Number
Hospitals	5,397
Dispensaries	13,646
Homes for people with leprosy	468
Homes for elderly, chronically ill or disabled	16,263
Orphanages	8,138
Crèches	10,352
Marriage counselling centres	10,624
Social rehabilitation centres	3,194
Other institutions	33,028
Total	**101,110**

Source: Agenzia Fides, Catholic Church Statistics 2026. School totals are my sums of Fides's figures.

Africa now has the most pupils in Catholic primary schools, more than 21 million. Most of the growth in Catholic numbers is in Africa.

> **What I could not find.** The Church owns land and buildings in almost every country, but no official global figure exists for its land holdings or its carbon footprint. Most property belongs to dioceses, religious orders and charities, each a separate legal owner, not to the Vatican. Totals sometimes quoted online have no verifiable source, so I have left them out.

5. The Vatican offices that do this work

The Holy See (the Church's central government) and Vatican City State (the territory) are legally separate. Science, economics and ecology are spread across several offices:

TABLE 4. WHO DOES WHAT

Office	Role in science, economy and environment
Dicastery for Promoting Integral Human Development	Set up by Pope Francis on 17 August 2016 by merging four pontifical councils, including Justice and Peace. It covers the economy and work, care for creation, health, migration and humanitarian emergencies, and develops the Church's social teaching. Runs the Laudato Si' Action Platform and has a Commission for Ecology.
Pontifical Academy of Sciences	Independent scientific academy of about 80 members (section 1). Holds study weeks and workshops; co-authored the 2015 climate statement.
Pontifical Academy of Social Sciences	Founded by John Paul II in 1994 to study the social, economic, political and legal sciences in the light of Church teaching. Published *Mensuram Bonam* (2022).
Vatican Observatory (Specola Vaticana)	Astronomical research from Castel Gandolfo and Tucson, Arizona; runs the VATT; summer schools and education.
Secretariat of State	Signs treaties for the Holy See; joined the UN climate convention and the Paris Agreement in 2022; represents the Holy See at COPs. Worked with the Governorate to measure emissions and prepare the NDCs.
Governorate of Vatican City State	Runs the state itself: buildings, energy, vehicles, waste. Its plan "Ecological Conversion 2030" (2023) sets out the state's carbon-neutral projects. Co-leads the agrivoltaic plant.
Administration of the Patrimony of the Apostolic See (APSA)	Manages the Holy See's property and investments; co-leads the agrivoltaic plant with the Governorate.
Laudato Si' Higher Education Centre / Borgo Laudato Si'	Created by Pope Francis in 2023; runs the ecology training centre, farm and gardens at Castel Gandolfo, opened by Leo XIV on 5 September 2025.

6. The Vatican's own footprint

Land and property

Vatican City State covers 0.44 km^2 and has 887 residents, plus about 6,000 people who work there daily. It also holds extraterritorial sites in Italy, including the papal estate at Castel Gandolfo and Santa Maria di Galeria (about 430 hectares), where Vatican Radio's transmitters have stood since 1957. Through APSA, the Holy See manages 4,281 property units in Italy (92% of the floor area in the province of Rome) and about 1,200 abroad, mainly in London, Paris, Geneva and Lausanne. In 2024 APSA paid more than €9 million in Italian property and corporate taxes.

Carbon

The Vatican's greenhouse gas inventory, reported to the UN, is small and dominated by road vehicles:



Source: Vatican City State First Biennial Transparency Report (Dec 2024); NDC (Sept 2025), base year restated as 19,300 t.

TABLE 5. WHERE THE VATICAN'S 2022 EMISSIONS COME FROM

Source	t CO_2e	Share
Transport (petrol and diesel vehicles)	11,250	71%
Heating (gas, district heating)	3,460	22%
Refrigerant gases (HFCs)	1,180	7%
Total	**15,890**	**100%**

For scale, the Vatican's emissions are 0.0000432% of the world's (2022). The state produces no fossil fuels and almost no energy of its own. All its electricity is imported, and the transparency report says it is all from renewable sources. Electricity use fell about 20% between 2019 and 2022. Gas burned at the Vatican Museums' heating plant fell 47% between 2016 and 2022.

Targets

- **2020:** Pope Francis committed Vatican City State to net zero by 2050.
- **May 2023, first NDC:** 20% below 2011 by 2030.
- **September 2025, second NDC:** at least 28% below 2011 by 2035. 60 vehicle charging points are planned by 2030.

Energy projects

- **2008:** about 2,400 solar panels on the roof of the Paul VI Audience Hall, under Benedict XVI.
- **2023:** the Governorate's "Ecological Conversion 2030" plan and the start of an all-electric fleet.
- **December 2024:** a solar array at the Cortile delle Corazze, the entrance to the Vatican Museums.
- **The big one, Santa Maria di Galeria:** in the motu proprio *Fratello Sole* (Brother Sun, 21 June 2024), Pope Francis ordered an agrivoltaic plant, with solar panels above working farmland, to power Vatican Radio and the whole of Vatican City State. Italy and the Holy See signed a treaty for it on 31 July 2025. Italy ratified it by Law 52 of 1 April 2026, and it came into force on 27 May 2026. Rome's utility ACEA has since signed a protocol to help build it. Reports put it at 80–90 MW, about €100 million and 200 hectares. Power not needed by the Vatican may be supplied to other Holy See properties or the Italian grid.

7. What businesses here can take from this

- **Measure it, however small.** The world's smallest state publishes an emissions inventory with a base year and two dated targets. Most NI manufacturers are bigger emitters than the Vatican and publish nothing. CBAM and CSRD customers will soon ask.
- **Find your biggest source.** For the Vatican it is vehicles (71%), not buildings. For most firms the biggest source is not where they expect.
- **Land can do two jobs.** Agrivoltaics, panels above grazing or crops, is worth a look for NI farms and food processors with land and big electricity bills.
- **Values-led money is real money.** Church investors, Catholic schools and hospitals buy goods and services. Guidance such as *Mensuram Bonam* shapes who they buy from and invest in.
- **Long-term institutions think in centuries.** An observatory that grew out of a 1582 calendar reform, and an academy dating from 1603, show the value of building institutions that outlast their founders.

Sources

1. Agenzia Fides, Catholic Church Statistics 2026 (16 Oct 2026). https://fides.org/en/stats
2. Vatican City State, First Biennial Transparency Report to the UNFCCC (Dec 2024). https://unfccc.int/sites/default/files/resource/Vatican%20City%20State%20BTR%20-%20December%202024.pdf
3. Vatican City State, Nationally Determined Contribution (Sept 2025). https://unfccc.int/sites/default/files/2025-09/Holy%20See_NDC%202035_September%202025.pdf
4. Vatican City State, Nationally Determined Contribution (May 2023). https://unfccc.int/sites/default/files/NDC/2023-05/Vatican%20City%20State%20NDCs%20-%20May%202023.pdf
5. Vatican City State, First National Communication to the UNFCCC (2025). https://unfccc.int/sites/default/files/resource/HolySeeNationalCommunication01.pdf
6. National Catholic Reporter, Vatican submits new climate target ahead of COP30 (Oct 2025). https://www.ncronline.org/vatican/vatican-news/vatican-submits-new-climate-target-ahead-cop30-un-summit
7. Vatican News, Pope orders construction of agrivoltaic plant for the Vatican (26 Jun 2024). https://www.vaticannews.va/en/pope/news/2024-06/pope-orders-construction-of-agrivoltaic-plant-for-the-vatican.html
8. Italian Ministry of Foreign Affairs, Holy See: signing of bilateral agreement on agrivoltaic system (31 Jul 2025). https://www.esteri.it/en/sala_stampa/archivionotizie/retediplomatica/2025/07/santa-sede-firma-accordo-bilaterale-impianto-agrivoltaico/
9. Italian Chamber of Deputies, ratification of the Italy–Holy See agrivoltaic agreement (A.C. 2759). https://temi.camera.it/leg19/provvedimento/ratifica-ed-esecuzione-dell-accordo-tra-la-repubblica-italiana-e-la-santa-sede-per-un-impianto-agrivoltaico-a-santa-maria-di-galeria-fatto-a-roma-il-31-luglio-2025.html
10. QualEnergia, The Vatican plans an 80–90 MW agrivoltaic plant near Rome (2026). https://www.qualenergia.it/articoli/vaticano-agrivoltaico-80-90-mw-porte-roma/
11. Affaritaliani, ACEA signs protocol for the agrivoltaic plant at Santa Maria di Galeria (2026). https://www.affaritaliani.it/economia/notizie-aziende/acea-firmato-il-protocollo-per-limpianto-agrivoltaico-nellarea-vaticana-di-santa-maria-di-galeria.html
12. Sky TG24, APSA 2024 accounts (29 Jul 2025). https://tg24.sky.it/economia/2025/07/29/utili-vaticano-papa-leone-xiv-
13. InfoVaticana, APSA report 2025 (4 Aug 2026). https://infovaticana.com/it/2026/08/04/rapporto-apsa-2025-il-patrimonio-netto-della-santa-sede-ammonta-a-2686-milioni-di-euro/
14. Dicastery for Promoting Integral Human Development, The Dicastery. https://www.humandevelopment.va/en/il-dicastero.html
15. Pope Francis, Statutes of the Dicastery for Promoting Integral Human Development (2016). https://www.vatican.va/content/francesco/en/motu_proprio/documents/papa-francesco_20160817_statuto-dicastero-servizio-sviluppo-umano-integrale.html
16. Vatican News, Laudato Si' Action Platform launches Digital Atlas; five-year report (Nov 2026). https://www.vaticannews.va/en/vatican-city/news/2026-11/laudato-si-platform-5-year-report-digital-atlas-press-conference.html
17. Pope Francis, Laudato si' (2015). https://www.vatican.va/content/francesco/en/encyclicals/documents/papa-francesco_20150524_enciclica-laudato-si.html
18. Pope Francis, Laudate Deum (2023). https://www.vatican.va/content/francesco/en/apost_exhortations/documents/20231004-laudate-deum.html
19. Pope Leo XIV, Homily at the Mass for the Care of Creation (9 Jul 2025). https://www.vatican.va/content/leo-xiv/en/homilies/2025/documents/20250709-omelia-custodia-creazione.html
20. Aleteia, Pope Leo XIV opens Borgo Laudato Si' in Castel Gandolfo (6 Sep 2025). https://aleteia.org/2025/09/06/pope-leo-xiv-opens-borgo-laudato-si-in-castel-gandolfo/
21. Pontifical Academies of Sciences and Social Sciences, Climate Change and the Common Good (Apr 2015). https://dcmpx.remotevs.com/org/documentcloud/s3/PL/documents/2068632/climate-change-and-the-common-good.pdf
22. IIASA, Vatican climate symposium and declaration (29 Apr 2015). https://iiasa.ac.at/web/home/about/news/150429-Vatican.html
23. Irish Catholic Bishops' Conference, Statement by Bishop William Crean on divestment from fossil fuels (24 Aug 2018). https://www.catholicbishops.ie/2018/08/24/statement-by-bishop-william-crean-announcing-the-decision-of-the-bishops-conference-to-divest-from-fossil-fuels/
24. National Catholic Reporter, Ireland's bishops pledge to divest from fossil fuels (Aug 2018). https://www.ncronline.org/print/pdf/node/171048
25. Pope Leo XIII, Rerum novarum (1891). https://www.vatican.va/content/leo-xiii/en/encyclicals/documents/hf_l-xiii_enc_15051891_rerum-novarum.html
26. Bishops' Conference of England and Wales, Magnifica humanitas to be published on 25 May (2026). https://www.cbcew.org.uk/pope-leo-xivs-first-encyclical-magnifica-humanitas-to-be-published-on-25-may/
27. PBS NewsHour / AP, Pope calls for robust regulation of AI (25 May 2026). https://www.pbs.org/newshour/world/pope-calls-for-robust-regulation-of-ai-in-manifesto-that-ponders-the-future-of-humanity
28. Pontifical Academy of Social Sciences, Mensuram Bonam (Nov 2022), as summarised by CCLA. https://www.ccla.co.uk/documents/catholic-faith-based-investing/download

29. Trenton Monitor / CNS, Vatican offers guidelines for investing in a way consistent with faith (28 Nov 2022). https://trentonmonitor.com/news/2022/nov/28/vatican-offers-guidelines-for-investing-in-a-way-c
30. Encyclopedia.com, Dictionary of Scientific Biography: Luca Pacioli. https://www.encyclopedia.com/doc/1G2-2830903265.html
31. The Guardian (via PressReader), The 70,000 workers showing the world another way to earn a living (25 Apr 2024). https://www.pressreader.com/australia/the-guardian-australia/20240425/281998972518118
32. Cardijn Research, José María Arizmendiarrieta. https://www.josephcardijn.com/fr/item/213
33. Vatican Observatory, Press kit and Media kit 2026. https://www.vaticanobservatory.org/wp-content/uploads/2025/12/VO-Media-Kit-2026a.pdf
34. Vatican Observatory Foundation, History. https://www.vaticanobservatory.org/?p=69
35. Nature, The Pontifical Academy of Sciences (1937). https://www.nature.com/articles/140965a0
36. History of Information, Mendel's discovery of the Mendelian ratios. https://historyofinformation.com/detail.php?entryid=3571
37. Revista Pesquisa FAPESP, The legend of an invisible monk (Mendel). https://revistapesquisa.fapesp.br/en/the-legend-of-an-invisible-monk/
38. Vatican Observatory, G. Coyne SJ, The Church's most recent attempt to dispel the Galileo myth. https://www.vaticanobservatory.org/wp-content/uploads/2017/01/Galileo_ed._McMullin.pdf

© 2026 Robert J. Park. Environment · Carbon · Public Affairs, Larne, Northern Ireland. robertjpark.co.uk. This briefing summarises public sources in our own words and is not legal, financial or investment advice.